Question	Answer	Marks
12(a)(i)	$I = I_0 e^{-\mu x}$	C1
	$= I_0 \exp(-0.90 \times 2.8)$ $= 0.080\, I_0$	A1
12(a)(ii)	$I = I_0 \exp[(-0.90 \times 1.5) \times (-3.0 \times 1.3)]$	C1
	$= I_0\ (0.259 \times 0.20)$ $= 0.0052\, I_0$	A1
12(b)(i)	difference in degrees of blackening	M1
	between structures	A1
12(b)(ii)	large difference in intensities so good contrast	B1

Question	Answer	Marks
13(a)	emission of particles/radiation by unstable nucleus	B1